

GREEN ALIBI — Project Report

1. Project Overview

GREEN ALIBI is an independent satellite-based verification study examining whether Solar-Induced Fluorescence (SIF) — a physical proxy for photosynthetic activity — registers vegetation and crop drought stress earlier than the Normalized Difference Vegetation Index (NDVI), the reflectance-based greenness metric that underlies much of India's current drought-monitoring and crop-insurance assessment framework. The study is situated in the Marathwada region of Maharashtra, an area with a well-documented recent history of recurring agricultural drought, and evaluates satellite data across eight growing seasons: 2015, 2016, 2017, 2018, 2019, 2020, 2022, and 2023. The study originally covered three years (2015, 2018, 2020); five more (2016, 2017, 2019, 2022, 2023) were added in a later expansion pass specifically to address this project's own recurring small-sample limitation, using the identical acquisition and processing pipeline throughout so all eight years are directly comparable.

2. Scientific Rationale

NDVI is a reflectance-based index, derived from how much visible and near-infrared light a leaf's surface reflects. Reflectance changes only once a plant's internal structure has already begun to visibly degrade — thinning leaves, reduced chlorophyll content, canopy stress. SIF is grounded in a more direct physical mechanism: when a leaf absorbs sunlight for photosynthesis, a small fraction of the absorbed energy is not converted into chemical energy but is instead re-emitted as fluorescence, a process governed by the quantum yield of photosynthesis. Under physiological stress, photosynthetic efficiency declines before outward greenness does, and this is expected to register as a measurable change in fluorescence emission ahead of any NDVI decline. NDVI, in this framing, reflects a plant's outward appearance, while SIF reflects the underlying physiological state.

India's drought declaration process, and the crop-insurance payout mechanism under the Pradhan Mantri Fasal Bima Yojana (PMFBY) that depends on it, relies substantially on rainfall-deficit and NDVI-based assessment — both indicators that are known to respond only after visible crop stress has already set in. Delayed drought recognition translates directly into delayed insurance payouts, at the point when affected farmers most need them. This project treats the existence and size of a SIF-NDVI lag as an empirical, testable question rather than an assumption, with the intent of establishing whether such a lag — if present and sufficiently large — could constitute a physics-based case for an earlier-warning indicator than those currently used in policy.

3. Research Questions and Hypotheses

Research Questions:

RQ1: Does SIF decline measurably earlier than NDVI during documented drought-onset periods in the study region?

RQ2: Is the SIF-NDVI lag, where it exists, consistent across different districts, or does it vary meaningfully by geography?

RQ3: How does the timing of SIF-based stress onset compare against official drought-declaration timelines for the same districts and years?

RQ4: Does the observed SIF-NDVI lag translate into a practically meaningful early-warning window relevant to crop-insurance and drought-relief timelines?

Hypotheses:

H1: SIF declines significantly before NDVI during a given drought episode, consistent with fluorescence capturing a drop in photosynthetic efficiency ahead of any visible, reflectance-based greenness loss.

H2: The magnitude and timing of SIF-NDVI divergence is not uniform across the study region, varying with local geography and rainfall regime.

H3: Drought severity, as measured independently through rainfall deficit, amplifies the size of the SIF-NDVI lag.

4. Study Region and Data Sources

The study region comprises the eight districts constituting Marathwada — Chhatrapati Sambhajnagar (Aurangabad), Jalna, Parbhani, Hingoli, Nanded, Beed (Bid in the FAO GAUL dataset used for boundary definition — see Section 5.3), Latur, and Dharashiv (Osmanabad) — selected for its recurring, well-documented drought history. Eight growing seasons were analyzed: June 1 through late December (day-of-year 153–361) of 2015, 2016, 2017, 2018, 2019, 2020, 2022, and 2023.

Four independent datasets were used:

- **SIF:** GOSIF v2, a global 0.05° 8-day solar-induced fluorescence product (Li & Xiao, 2019), distributed by the Global Ecology Group at the University of New Hampshire.
- **NDVI:** MODIS MOD13Q1, a 16-day, 250 m Vegetation Index product incorporating a Maximum-Value-Composite cloud-suppression algorithm, accessed via Google Earth Engine.
- **Land cover:** MODIS MCD12Q1 (IGBP classification), used to construct a cropland mask (classes 12 and 14) restricting analysis to agricultural pixels.
- **Precipitation:** CHIRPS daily rainfall, accessed via Google Earth Engine, used both for all eight study years and for a twenty-year (2001–2020) climatological baseline.

5. Methodology

5.1 SIF Acquisition and Processing

Two hundred and sixteen 8-day GOSIF GeoTIFF files (twenty-seven per study year, across eight years) were acquired for the June–late-December window. The original three years' files were pulled first; the five additional years' files (135 total) were downloaded in a later expansion pass using the identical seasonal window and file-naming convention, so the existing clipping script required no changes to pick them up. Each file was clipped to the Marathwada study region, with a scale factor of 0.0001 applied to convert raw digital values to physical SIF units, and GOSIF's fill-value codes (32766 for water, 32767 for non-vegetated or missing data) masked prior to any averaging. A regional daily/8-day mean SIF value was then computed for each date, alongside a valid-pixel-fraction quality metric.

5.2 NDVI Acquisition and Processing

NDVI was extracted via Google Earth Engine using MOD13Q1, filtered by its native SummaryQA band (retaining only good- and marginal-quality pixels) and restricted to cropland pixels via the MCD12Q1-derived mask. An initial extraction using raw MOD09Q1 reflectance without quality filtering produced a physically implausible time series (single-period swings inconsistent with vegetation phenology), traced to cloud and cloud-shadow contamination during the monsoon-season observation window; the pipeline was rebuilt around the cloud-screened MOD13Q1 product to resolve this. The five additional years added in the sample-size expansion were extracted using this same MOD13Q1/SummaryQA/cropland-mask pipeline, run via a Python script (`src/acquisition/gee_data_acquisition.py`) rather than interactively.

5.3 Boundary Definition

The initial study-region boundary was defined as a rectangular bounding box (75.0–78.5°E, 17.5–20.5°N). Subsequent visual verification in Google Earth Engine identified that this rectangle extended beyond Marathwada into neighboring Telangana districts (Nizamabad, Bidar, Adilabad) and into Solapur and Yavatmal, both outside the study region and subject to different rainfall regimes. The boundary definition was corrected to the precise union of the eight Marathwada districts, sourced from the FAO GAUL 2015 level-2 administrative boundary dataset. A naming-convention discrepancy (Beed district being stored under the earlier spelling “Bid” in the 2015-vintage dataset) was identified and corrected in the original three-year run. During the sample-size expansion, this exact same naming mismatch reappeared in the separate district list used by the newly Python-scripted acquisition step, and silently dropped Bid from the merged region boundary across all eight years' region-level and district-level exports the first time that script was run. It was caught by counting rows in the by-district output (56 rather than the expected 64) rather than by any error, fixed, and the full eight-year extraction was re-run and re-verified before any figure in

this report was computed.

5.4 Temporal Alignment and Lag Calculation

SIF (8-day) and NDVI (16-day) series were merged per year using nearest-date matching. A quantitative lag metric was constructed by computing, for each year, the day-of-year at which each series (normalized 0–1 within year) crossed a series of decline thresholds (90%, 80%, 70%, 60%, and 50% of seasonal peak), via linear interpolation between observed dates. The lag at each threshold was defined as the difference between the NDVI crossing date and the corresponding SIF crossing date.

As a robustness check on this method, a second, independent lag estimate was computed via time-lagged cross-correlation — interpolating both series onto a common daily grid and finding the time-shift that maximizes their correlation, rather than relying on threshold crossings. Across all eight years, this method never favors NDVI leading SIF, but only finds a clearly positive lag in half of the eight years; the other half, including 2018, are best aligned at zero lag. This divergence from the threshold-crossing method's exact numbers is reported directly in Section 7 rather than resolved in favor of either method.

5.5 Spatial Analysis

District-level and pixel-level spatial comparisons were constructed for a representative day-of-year (273) across all eight study years, using the precise administrative boundary for masking. Per-district zonal statistics (mean SIF, mean rainfall anomaly) were computed by masking each dataset to individual district polygons.

5.6 Rainfall Validation

CHIRPS precipitation totals for the June–December window were computed for all eight study years and for a twenty-year (2001–2020) climatological period, establishing a measured baseline against which each study year's rainfall anomaly (percentage departure, z-score) was calculated. A year is classified as a drought year if its anomaly z-score falls below -0.5 — the threshold implied by, and made explicit from, the original study's own two-drought/one-normal classification. Applied to all eight years, only 2015 and 2018 qualify; the five newly added years all fall within roughly one climatological standard deviation of normal. This was extended to the district level, using the region-wide climatological mean as a shared reference baseline across all eight districts.

5.7 Spatial Correlation Testing

District-level mean SIF (Section 5.5) and district-level rainfall anomaly (Section 5.6) were merged on district and year, producing 64 district-year observations ($8 \text{ districts} \times 8 \text{ years}$), up from the original 24 ($8 \text{ districts} \times 3 \text{ years}$). Pearson and Spearman correlation coefficients were computed between the two variables to test, quantitatively, whether the spatial correspondence visible in the mapped output was a genuine statistical relationship rather than a visual impression alone.

5.8 Uncertainty Quantification, Spatial Autocorrelation, and Official-Declaration Comparison

Going back through this project critically, three further checks were added, and all three were re-run against the expanded eight-year dataset. First, a case-resampling bootstrap (2,000 replicates per year) placed a 95% confidence interval around each year's cross-correlation lag estimate, resampling which of each year's satellite overpass observations were represented in the fit. Second, Moran's I was computed for district-level mean SIF and rainfall anomaly in each study year, using a Queen-contiguity spatial weights matrix built from the actual district polygons, to quantify — rather than only assert — the spatial non-independence flagged as a limitation in Section 5.7. Third, the SIF and NDVI decline-onset dates for 2018 were compared against the one well-documented official Maharashtra government drought-declaration date I could find (31 October 2018), to address RQ3's original, more policy-relevant framing; re-running this comparison against the corrected eight-district boundary changed which index (SIF or NDVI) crosses first in 2018 specifically, detailed in Section 7. All three are detailed in Section 7 (Key Findings) and Section 8 (Limitations).

6. Development Process and Quality Assurance

Several methodological issues were identified and resolved over the course of the analysis, each

addressed before proceeding rather than deferred:

- An initial NDVI extraction lacked cloud/cloud-shadow quality filtering, producing an implausibly noisy time series; resolved by switching from raw MOD09Q1 reflectance to the cloud-screened MOD13Q1 product with explicit QA-band masking.
- An Earth Engine geometry-transform error, caused by clipping a multi-resolution composite image, was resolved by removing a redundant clip operation, since region-restriction was already handled by the reducer's geometry parameter. This same class of error reappeared, in a different form, when the acquisition step was later translated into a standalone Python script for the sample-size expansion (Section 5.2): clipping a MODIS image on its native sinusoidal projection against an EPSG:4326 boundary polygon threw a transform error for certain district edge geometries, resolved the same way, by removing the redundant clip and relying on the reducer's own geometry restriction.
- A plot-labeling error incorrectly identified 2018 as a normal monsoon year rather than a drought year; corrected via explicit set-membership logic rather than single-year conditional checks. Every drought-year classification used throughout this project, including after the eight-year expansion, is now derived from the actual measured rainfall anomaly (Section 5.6) rather than hardcoded per year.
- An initial visual assessment of the SIF-NDVI lag suggested a materially larger lag in drought years than in the normal year; on rigorous recalculation using a percentage-of-peak threshold method, this specific claim was not supported and was withdrawn in favor of the more limited, defensible finding that SIF's decline precedes NDVI's decline generally. This finding has since replicated at eight years: the two years now meeting this study's drought threshold still show a smaller mean lag than the six normal years.
- The initial study-region boundary (a rectangular bounding box) was found, on visual inspection, to include areas outside Marathwada with differing rainfall regimes; the boundary was corrected to a precise administrative polygon (FAO GAUL level-2), with a naming-convention gap (Beed vs. "Bid") identified and corrected via an explicit feature-count verification step added to the extraction pipeline. This exact gap reappeared independently in the Python-scripted acquisition step written for the sample-size expansion (Section 5.3) and was caught the same way, by actually counting rows in the output rather than assuming a clean run meant a correct one.
- Following the boundary correction, a spatial verification map that appeared visually unchanged prompted a dedicated diagnostic check (overlaying the true boundary polygon directly on the raster output) to confirm, independently of visual impression, that the underlying masking had in fact been corrected; this confirmed the raster data was correctly masked, and identified the visualization script itself — not the underlying data — as the remaining unupdated component.
- A rainfall-validation step confirmed the drought/normal classification used throughout the study via independently measured precipitation deficit. Extended to eight years, this step also revealed that only two of the eight years (2015, 2018) actually meet this study's own drought threshold — the original three-year sample, being two-thirds drought years, was considerably more drought-heavy than the region's typical eight-year climate record (one-quarter drought years).
- The spatial correspondence between SIF and rainfall deficit had initially been reported only as a visual, physically-motivated consistency check. This was subsequently tested formally: a Pearson and Spearman correlation across the original 24 district-year observations returned $r = 0.837$ and $\rho = 0.857$. Re-run across the expanded 64 district-year observations, this dropped to $r = 0.567$ and $\rho = 0.551$ — still highly significant ($p < 0.0001$), but a substantially weaker relationship, consistent with the original correlation having been computed from an unusually drought-heavy, more linearly-behaved sample.
- An initial attempt to quantify uncertainty in the cross-correlation lag estimates used a moving-block bootstrap applied directly to the interpolated daily SIF/NDVI series. Every year's confidence interval collapsed to a single spurious point at lag 0 — block-shuffling a strongly non-stationary decline curve destroys the trend that produces the lag signal, rather than measuring genuine sampling uncertainty. This was replaced with a case-resampling bootstrap over the original sparse satellite observation dates, which preserves the decline curve's shape in every replicate and produced sensible, non-degenerate confidence intervals for all eight years (GA_Development_Log.md has the full account).
- A locatable, single official drought-declaration date for any study year besides 2018 could not be found despite a dedicated search; 2015 Marathwada drought coverage in public sources, for example, concentrates on the subsequent, better-documented 2016 water-scarcity crisis rather than a single dated Kharif declaration. Rather than approximate or infer a date for another year, the official-declaration comparison (Section 5.8) remains reported for 2018 only, with this gap stated explicitly rather than silently worked around.

7. Key Findings

1. SIF's post-peak seasonal decline precedes NDVI's decline in seven of the eight study years (2015, 2016, 2017, 2019, 2020, 2022, 2023), supporting the core premise that SIF is a more temporally responsive indicator of physiological change than NDVI. 2018 is a genuine, replicated exception — its mean lag is marginally negative rather than positive — discussed directly in Finding 4 rather than averaged away.
2. The hypothesis that drought conditions amplify the SIF-NDVI lag (H3) is not supported by this dataset. Mean lag across resolved thresholds ranged from -1.1 days (2018, drought) to 23.8 days (2016, normal); the two years meeting this study's drought threshold averaged a smaller lag (7.6 days) than the six normal years (15.0 days) — the same direction found in the original three-year study, now confirmed at more than double the sample size.
3. Only two of the eight study years — 2015 and 2018, the same two years the original study called drought years — actually meet this study's own quantitative rainfall-anomaly drought threshold (anomaly z-score below -0.5 relative to the 2001–2020 climatology). The remaining five newly added years (2016, 2017, 2019, 2022, 2023) all fall within roughly one standard deviation of normal. This means the original three-year sample was considerably more drought-heavy (two of three years) than Marathwada's actual eight-year climate record (two of eight years) — a finding about sample representativeness independent of anything about the SIF-NDVI relationship itself.
4. 2018 is a consistent exception to Finding 1 across every method used in this study: its threshold-crossing lag is marginally negative (Finding 2), its cross-correlation-maximizing lag is exactly zero (Finding 7), its bootstrap confidence interval shows only 8.1% of replicates favoring a positive lag (Finding 8), and — after the boundary correction described in Section 5.3 — its official-declaration comparison (Finding 10) now shows NDVI crossing its decline threshold before SIF rather than after, the opposite of what the original three-year study reported. This study does not have the crop-type, sowing-date, or within-season rainfall-timing records that would explain why 2018 specifically differs from the other seven years, and this gap is reported directly rather than guessed at.
5. District-level spatial analysis identifies Osmanabad as the lowest-mean-SIF district in all eight years studied — a pattern that held in the original three years and has now replicated across the full eight-year record. The finding that Nanded is consistently the highest-SIF district does not survive the expanded sample: Aurangabad is highest in four of the eight years, Nanded in three, and Jalna in one.
6. Rainfall deficit was found to be spatially uneven within Marathwada during both drought years, particularly in 2018, where the eastern districts (Nanded, Hingoli) recorded near-normal or above-normal rainfall despite a region-wide deficit of 18.3%. This correspondence was confirmed statistically: mean SIF and rainfall anomaly are significantly correlated across all 64 district-year observations (Pearson $r = 0.567$, Spearman $\rho = 0.551$, both $p < 0.0001$) — real and highly significant, but substantially weaker than the $r = 0.837$ reported from the original, smaller, more drought-heavy sample (Finding 3).
7. A cross-correlation-based lag estimate, computed independently of the threshold-crossing method, confirms a weaker but still real version of Finding 1: across all eight years, the correlation-maximizing lag is never negative (never favoring NDVI leading SIF), but is clearly positive in only four of the eight years (2015, 2016, 2017, 2019); the other four, including 2018, show no detectable lag either direction. This does not reproduce the same inter-annual ranking as the threshold-crossing method (Finding 2) — both methods agree only that 2018 has the smallest or most negative lag of any year.
8. A bootstrap confidence interval around each year's cross-correlation lag estimate confirms Finding 7's direction with high confidence — the lag was at or above zero in 100% of resampled replicates for every one of the eight years — but shows the exact magnitude is considerably less certain than the point estimates suggest, with every one of the 28 possible pairwise between-year comparisons statistically indistinguishable.
9. Moran's I confirms that rainfall anomaly is significantly spatially clustered in all eight study years, but mean SIF is significantly clustered in only four of the eight (2015, 2016, 2018, 2020) — a more qualified result than the original three-year study, where all three years were significant for both variables. This suggests SIF's spatial clustering may itself strengthen during drought or near-drought years and weaken during more ordinary ones, a genuinely new finding from the expanded sample.
10. Comparing SIF's and NDVI's 2018 decline-onset dates against the one well-documented official government drought declaration available (31 October 2018) shows both satellite indicators crossed their 90%-decline threshold roughly seven to eight weeks earlier (NDVI: 4 September, 57 days prior; SIF: 8 September, 53 days prior) — a substantially larger gap than the small SIF-versus-NDVI difference in either direction. Notably, this specific ordering (NDVI first, then SIF) is the reverse of what the original three-year study reported for 2018, a direct

consequence of the boundary correction described in Section 5.3, and is consistent with Findings 2, 4, and 7 all identifying 2018 as this study's one year where SIF does not clearly lead.

8. Limitations

- The GOSIF product used for SIF is a statistically modeled reconstruction, not a direct satellite retrieval, built from OCO-2 SIF soundings combined with MODIS reflectance data and meteorological reanalysis. Since MODIS reflectance also underlies the NDVI product used for comparison, the two variables are not fully independent at the input-data level — an inherent property of the dataset choice rather than a design flaw, but one this study does not attempt to quantify or correct for.
- The sample now covers eight years, up from the original three, specifically to address this study's own recurring small-sample limitation. Even at eight years, only two (2015, 2018) meet this study's own drought threshold, so the drought-versus-normal comparison remains a two-versus-six comparison rather than a balanced one, and a single additional future drought year could still shift the drought-group average considerably.
- 2018 is a consistent, multiply-confirmed exception to this study's central SIF-leads-NDVI finding, and this study does not have the crop-type, sowing-date, or within-season rainfall-timing data that would explain why.
- The observed spatial correspondence between rainfall deficit and SIF stress was tested via Pearson and Spearman correlation ($r = 0.567$, $\rho = 0.551$, both $p < 0.0001$) rather than left as a visual check alone; however, the 64 district-year observations are not fully independent, since the eight districts are geographically adjacent and share regional weather systems, so this should be read as corroborating evidence rather than a fully independent statistical confirmation. Moran's I quantifies this directly: significant for rainfall anomaly in all eight years, but for mean SIF in only four of eight, meaning the effective independent sample size behind this correlation varies by year.
- The low-SIF spatial zones identified have not been validated against ground-level crop-stress reporting; the observed pattern is consistent with known Marathwada drought geography but has not been independently confirmed.
- District-level rainfall anomaly was computed relative to a single region-wide climatological baseline rather than a per-district climatology, which was not computed due to the additional data-processing scope required.
- The Google Earth Engine queries used to acquire NDVI, the cropland mask, and rainfall data now run as a checked-in Python script (`src/acquisition/gee_data_acquisition.py`) rather than only interactively, and this script produced all eight years of NDVI and rainfall data used in this report, including the five newly added years. Running it against the real Earth Engine service surfaced two genuine bugs (a missing registered Cloud project, and the MODIS-projection clip error described in Section 6) that a purely interactive workflow could plausibly have also hit without a code review catching them — both are documented in `GA_Development_Log.md`.
- The official-declaration comparison (Section 5.8, Finding 10) remains available for 2018 only; a single, comparably verifiable declaration date for another study year could not be located, so this comparison should not be read as covering all eight study years — and, per Finding 10, its specific SIF-versus-NDVI ordering for 2018 changed after the boundary correction, which is itself a reason for caution about reading any single-year comparison as more stable than the sample size behind it supports.